Learning Outcomes & Educational Impact Report

Comparative Analysis of Pre- and Post-Learning Assessments for the Engineering Playground Platform

Report date: 10 October 2026

Evidence base: participant responses to the pre- and post-learning assessments

01 | Executive Summary

Among 11 participants with matched assessment records, mean performance increased from 5.18/15 before platform use to 13.64/15 afterwards. The average paired gain was 8.45 points, and all matched participants recorded a higher post-assessment score.

Pre-assessment responses	Post-assessment responses	Matched participants	Improved scores
12	12	11	100%

Key interpretation. The results provide encouraging descriptive evidence of improved knowledge of foundational electrical-circuit concepts following the learning experience. They do not, on their own, establish that the platform exclusively caused the improvement.

02 | Evaluation Design & Methodology

A descriptive pre/post design was used. Each assessment comprised 15 questions addressing related introductory electrical-circuit concepts, including energy sources, closed paths, current, resistance, Ohm's law, and series/parallel connections. Records were matched by email address solely for analysis; identifying information is excluded from this report.

- 12 pre-assessment responses and 12 post-assessment responses were available.
- 11 participants had scores in both assessments; 1 pre-only and 1 post-only records were excluded from paired comparisons.
- Individual learning gain was calculated as post-assessment score minus pre-assessment score; summary results use the mean of these paired differences.
- The primary comparison uses matched participants rather than the two complete, partially different response groups.

03 | Participant Profile

The pre-assessment captured participants' fields of study and self-reported prior familiarity with electrical circuits. These characteristics describe the pre-assessment sample, not necessarily all platform users.

Field of study	Participants	Share
Information Systems	5	41.7%
Computer Engineering	2	16.7%
Computer Science	2	16.7%
Electrical Engineering	1	8.3%

Management Information System	1	8.3%
Computer science	1	8.3%

Self-reported prior knowledge

Reported level	Participants
Basic knowledge	6
Moderate knowledge	3
No prior knowledge	3

04 | Paired Learning Outcomes

Pre-test mean /15	Post-test mean /15	Mean gain /15
5.18	13.64	+8.45

The mean score increased from 34.5% to 90.9%, a rise of 56.4 percentage points. The normalized gain, calculated against the remaining attainable score, was 86.1%. These metrics express different aspects of change and should not be treated as interchangeable.

Anonymous participant-level comparison

Participant ID	Pre /15	Post /15	Change
01	5	15	+10
02	1	15	+14
03	6	12	+6
04	4	14	+10
05	5	15	+10
06	11	13	+2
07	9	14	+5
08	3	15	+12
09	2	11	+9
10	5	14	+9
11	6	12	+6

Individual gains ranged from +2 to +14 points. Median scores were 5/15 before and 14/15 after the learning experience.

05 | All-Response Descriptive Results

Across all available responses, the pre-assessment mean was 5.08/15 (n=12), while the post-assessment mean was 13.67/15 (n=12). Because the complete groups are not perfectly matched, these overall means are supplementary descriptive figures rather than the primary estimate of within-participant change.

06 | Interpretation of Educational Impact

The consistent upward shift across matched participants is compatible with improved understanding of the concepts sampled by the assessments. Nevertheless, knowledge-test performance should not be interpreted as direct evidence of practical circuit-building proficiency, simulation quality, independent problem-solving transfer, or long-term retention. These outcomes require separate measurement.

07 | Methodological Considerations

- The matched sample is small, limiting generalizability to wider student populations.
- Without a control or comparison group, alternative explanations for score gains cannot be excluded.
- The response files do not document time spent on the platform, activities completed, or user-experience ratings.
- The two assessments cover related concepts, but equivalent difficulty has not been formally established.
- The analysis uses recorded assessment scores; an independent item-by-item validation of the original answer keys remains advisable.

08 | Conclusions & Recommendations

The assessment evidence indicates substantial improvement in recorded knowledge scores among participants who completed both measures. The findings support continued evaluation and refinement of Engineering Playground, while stronger research designs are needed to substantiate causal impact and broader educational effectiveness.

- Review assessment answer keys and establish equivalent test forms for future cohorts.
- Expand the participant sample and document platform exposure and completed learning activities.
- Add user-experience feedback, hands-on performance measures, and a delayed retention assessment.
- Consider a comparison group and repeat the evaluation across academic backgrounds.

Prepared as a descriptive evaluation using the supplied pre- and post-learning response records. No participant-identifying information is disclosed.